Project Proposal(OOP)

Title: Bank Account Management System

Problem Statement

The proposed Bank Account Management System will allow bank administrators to efficiently manage customer accounts and basic banking transactions. The system will maintain records of account holders and support operations such as creating new accounts, depositing money, withdrawing funds, transferring money between accounts, and viewing account details. By automating these processes, the system aims to reduce the complexity and potential errors associated with manual record keeping.

Target User:

Bank administrators or bank staff who are responsible for managing customer accounts and bank customers whose account information and transaction records are maintained within the system.

Core Features:

- Account Creation
- Deposit Money
- Withdraw Money
- Fund Transfer

OOP Concepts To Be Used:

- Encapsulation
- Inheritance
- Polymorphism
- Abstraction
- Exception Handling

Project Architecture Description:

The **Bank Account Management System** will follow a **modular architecture** consisting of different layers. The **model layer** will represent entities such as accounts and customers. The **service layer** will handle banking operations like account creation, deposit, withdrawal, and fund transfer. The **utility layer** will manage tasks such as file handling and data storage. Finally, the **main layer** will provide a menu-driven interface for user interaction. This architecture ensures clear separation of responsibilities and improves maintainability of the system.